

MINQIN, China

WMO: 526810

Lat: 38.6320N

Lon: 103.0890E

Elev: 1367

StdP: 85.95

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.5	-16.0	-27.6	0.4	-10.3	-25.9	0.4	-9.6	8.9	-1.5	7.7	-1.8	1.4	90	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	34.1	17.1	32.6	16.5	31.1	15.9	19.7	28.3	18.8	27.1	18.0	26.4	3.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	14.6	22.4	16.2	13.6	21.9	15.3	12.8	21.4	63.1	28.6	59.7	27.0	56.7	26.5	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	7.5	6.3	-21.3	37.6	2.6	2.1	-23.2	39.1	-24.7	40.3	-26.1	41.4	-27.9	42.9		
			-21.9	21.1	2.4	1.5	-23.6	22.2	-25.0	23.1	-26.3	23.9	-28.1	25.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-7.4	-2.8	4.4	12.1	17.7	22.3	24.3	22.8	17.3	10.0	1.8	-5.3
	DBStd	11.64	4.33	4.80	4.81	4.59	3.78	3.08	3.12	3.03	3.45	3.95	4.16	4.04
	HDD10.0	1884	539	359	183	31	2	0	0	0	1	49	248	473
	HDD18.3	3631	798	592	433	193	57	6	1	3	59	260	497	731
	CDD10.0	1814	0	0	8	93	239	369	442	397	218	48	1	0
	CDD18.3	521	0	0	0	4	37	126	185	142	27	1	0	0
	CDH23.3	5448	0	0	1	74	456	1326	1962	1347	269	12	0	0
	CDH26.7	2143	0	0	0	14	120	514	909	526	58	1	0	0
Wind	WSAvg	2.7	2.2	2.5	3.0	3.3	3.1	2.9	2.8	2.7	2.4	2.4	2.4	2.4
Precipitation	PrecAvg	111	1	1	3	5	10	14	20	31	16	8	2	0
	PrecMax	185	3	5	23	19	40	60	74	88	63	33	24	3
	PrecMin	42	0	0	0	0	0	2	3	3	1	0	0	0
	PrecStd	35	1	1	4	5	10	13	16	22	13	8	5	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	13.4	21.5	28.6	31.7	34.4	37.1	35.0	30.9	24.9	16.6	9.3
		MCWB	0.1	3.5	7.2	11.0	13.0	15.8	18.7	18.1	15.2	10.9	5.5	0.8
	2%	DB	4.6	10.0	18.1	25.4	29.7	32.6	34.9	33.0	28.3	22.2	14.1	6.0
		MCWB	-1.9	1.4	5.7	10.0	12.7	15.3	17.7	17.3	13.9	10.1	4.6	-0.9
	5%	DB	2.0	7.6	15.6	23.0	27.8	31.3	33.2	31.4	26.4	19.9	11.7	3.7
		MCWB	-3.5	0.2	4.7	8.9	11.9	14.9	17.2	16.7	13.5	9.0	3.3	-2.1
	10%	DB	-0.2	5.3	13.2	20.7	25.9	29.8	31.6	29.7	24.5	17.7	9.3	1.6
		MCWB	-4.8	-1.0	3.5	7.9	11.1	14.4	16.9	16.2	13.1	8.0	2.2	-3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	3.7	7.9	12.3	14.9	18.4	21.6	21.0	18.0	12.7	6.4	1.1
		MCDB	7.8	12.2	18.8	23.6	26.4	27.9	32.8	27.1	22.3	20.9	14.4	8.4
	2%	WB	-1.8	1.9	6.4	10.8	13.7	17.5	20.0	19.7	16.5	11.2	5.0	-0.7
		MCDB	4.3	9.2	17.0	23.0	26.1	27.2	29.6	27.6	23.2	18.9	12.9	5.7
	5%	WB	-3.4	0.4	5.1	9.7	12.8	16.6	19.1	18.9	15.3	10.0	3.7	-2.0
		MCDB	1.7	7.0	14.7	21.0	24.9	26.1	27.8	26.8	21.7	18.1	10.7	3.4
	10%	WB	-4.7	-0.9	3.9	8.5	11.9	15.8	18.2	18.0	14.4	8.9	2.6	-3.2
		MCDB	-0.4	5.0	12.6	19.2	23.5	25.4	26.9	26.1	21.2	16.4	9.0	1.3
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.0	13.1	12.9	12.6	12.0	11.6	11.4	11.6	12.9	12.2	11.6
		MCDBR	14.8	15.9	16.4	16.0	15.3	14.4	14.2	13.9	14.8	15.9	15.4	13.9
		MCWBR	9.9	9.7	8.4	7.0	5.7	4.9	4.4	4.3	4.9	6.7	8.5	8.9
	5% WB	MCDBR	14.6	15.2	15.4	14.0	13.1	11.5	11.2	11.4	11.1	14.1	14.3	13.5
		MCWBR	9.9	9.3	8.0	6.3	5.3	4.2	4.1	4.0	4.5	6.5	8.1	8.8
Clear-Sky Solar Irradiance	taub	0.338	0.394	0.458	0.501	0.504	0.497	0.487	0.484	0.466	0.420	0.360	0.330	
	taud	2.066	1.933	1.796	1.736	1.791	1.871	1.970	1.961	1.935	1.966	2.053	2.087	
	Ebn at Noon	804	800	786	781	789	794	799	789	772	762	768	781	
	Edh at Noon	118	156	198	224	216	200	180	177	171	149	119	108	
All-Sky Solar Radiation	RadAvg	2.78	3.77	5.03	6.07	6.77	6.81	6.37	5.82	4.76	4.13	3.15	2.60	
	RadStd	0.28	0.20	0.29	0.30	0.28	0.43	0.38	0.36	0.37	0.23	0.19	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.53	N/A	N/A	+0.73	N/A	N/A	N/A	N/A	+124	+65
(47)	Regional (1 neighbor)	+0.36	N/A	N/A	+0.58	N/A	N/A	N/A	N/A	+124	+65

Nomenclature: See separate page